

# v4.1 — Preprocessing + QC Gate

**E2E outcome:** every ingested slide gets a per-slide fitted stain-normalization transform and an original  $\rightleftharpoons$  normalized viewer toggle; bad slides (blur, low tissue, faded stain) are blocked with a human-readable reason before any ML spend. This stage encodes the v1 lesson: garbage in  $\rightarrow$  hallucination out.

**Est:** 1 week. Depends on v4.0.

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## 1. Scope

**In:** Macenko stain fit + normalized 10 $\times$  layer, ICC handling, QC checks (tissue %, focus, pen/fold flags, stain sanity), QC review UI, first `awaiting_review` gate. **Out:** DICOM conversion (deferred  $\rightarrow$  v4.6), normalization above 10 $\times$  (see §2.3), any tumor detection.

## 2. Preprocess handler (`worker/preprocess.py`)

Runs after ingest; stage `preprocess`, no review gate (outputs feed QC which has the gate).

### 2.1 ICC

If the slide embeds an ICC profile (`openslide.color-profile` via `openslide-python`  $\geq 1.3$  or Pillow `ImageCms` on tiles): build a cached `ImageCms` transform to sRGB and apply to every tile read. If absent: assume sRGB, record `icc_applied: false` in output JSON. All downstream tile reads go through one function: `pipeline/tiles.py:read_region_srgb(slide, x_um, y_um, w_um, h_um, out_px)`.

### 2.2 Stain fit (Macenko, per-slide)

```
# pipeline/stain.py
```

```
from tiatoolbox.tools.stainnorm import MacenkoNormalizer
```

```
# 1. tissue mask at 1.25x: tiatoolbox OtsuTissueMasker on the thumbnail
```

```
# 2. sample up to 50 random 512x512 tissue patches at 10x (seeded RNG: seed = slide
checksum int)
```

```
# 3. reject patches with saturation-mean < 0.05 (background sneaking in)
```

# 4. fit ONE normalizer on the patch mosaic (concatenate to a single array)

# 5. persist stain matrix + max concentrations as JSON → `gcs .../preprocess/stain_params.json`

Target reference: fixed reference patch checked into `configs/stain_reference.png` (one well-stained TCGA patch — versioned; changing it bumps config hash and invalidates caches).  
Determinism: seeded sampling ⇒ identical stain params on re-run.

## 2.3 Normalized viewer layer

Generate a **normalized DZI pyramid only up to the 10× level** (~1.0 μm/px):

- Read 4096<sup>2</sup> regions at 10× via `read_region_srgb`, apply `normalizer.transform`, feed into a `pyvips` image assembled from memory buffers, `dzsave` → `pyramids/{slide_id}/norm/`.
- Viewer toggle: below 10× zoom OSD swaps tile layer to `norm`; beyond 10× it falls back to `orig` with a small "original colors" badge. Rationale: models consume normalized patches produced on-the-fly (40× mitosis patches normalized per-patch at inference in v4.3); a full 40× normalized pyramid would triple storage and add ~30 min/slide for a purely cosmetic depth.

## 2.4 Output JSON (`preprocess/output.json`)

```
{ "icc_applied": true, "stain_params_uri": "...", "norm_pyramid_uri": "...",  
  
  "thumbnail_uri": "...", "tissue_mask_uri": "...",  
  
  "tissue_area_mm2": 187.4, "model_versions": {"tiatoolbox": "1.6.0"} }
```

Tissue mask: PNG at 1.25×, 1 bit, stored — reused by every later stage.

## 3. QC handler (`worker/qc.py`)

Stage `qc`, ends in `awaiting_review` (warn) or `failed` (hard fail) or `done` (clean pass — auto-advance, no click needed).

### 3.1 Checks (all thresholds in `configs/qc.yaml`)

| Check           | Method                           | Default threshold           | Outcome   |
|-----------------|----------------------------------|-----------------------------|-----------|
| Tissue coverage | tissue mask area /<br>slide area | < 2% → fail; < 5% →<br>warn | fail/warn |

| Check        | Method                                                                                                                        | Default threshold                                                                    | Outcome   |
|--------------|-------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------|-----------|
| Focus        | variance of Laplacian (OpenCV, grayscale) per 512 <sup>2</sup> tile at 10×, on ≤ 400 sampled tissue tiles                     | tile blurry if VoL < 45; slide fail if > 30% blurry; warn if > 10%                   | fail/warn |
| Pen marks    | HSV gates (green/blue/black pen ranges, from HistoQC's defaults) on 1.25× thumbnail; connected components > 1 mm <sup>2</sup> | any → warn + region polygons                                                         | warn      |
| Folds        | high-saturation ∧ low-brightness ridges at 2.5×, morphological skeleton > 2 mm                                                | any → warn + polygons                                                                | warn      |
| Stain sanity | Macenko concentration stats: median H and E concentration on sampled patches                                                  | either < 0.15 (faded) or H:E ratio outside [0.3, 3.5] → warn; both degenerate → fail | fail/warn |

Calibrate thresholds once on ~20 known-bad + 50 known-good slides (you have archive access); record the calibration set ids in `configs/qc.yaml` comments.

### 3.2 Output JSON

```
{ "verdict": "warn",

  "checks": [

    { "name": "focus", "status": "warn", "metric": 0.13,

      "message": "13% of tissue tiles below sharpness threshold",

      "regions_um": [[[x,y],...], ...]},
```

...],

"config\_hash": "..."} }

### 3.3 Review semantics

- **fail** → case blocked; UI shows reason + "Request re-scan" (sets case status **needs\_rescan**, notifies technician role). Pathologist may **override a fail** (button + mandatory free-text justification → **review\_edits**, audit **score\_override**) — clinical reality: sometimes the only slide is a bad slide.
- **warn** → banner + flagged-region overlay (orange polygons); "Acknowledge and continue" = confirm.
- **pass** → auto-advance, silent.

## 4. Frontend

- Stage rail: QC entry shows verdict chip (green/orange/red).
- Overlay: QC region polygons rendered via the v4.0 **addOverlayLayer** API as SVG paths ( $\mu\text{m}$  → viewport via **umToViewport**); toggleable in a layer panel (this layer panel is reused by every later stage — build it generically now: **{id, label, color, visible}**).
- Original ⇌ normalized toggle in viewer toolbar.

## 5. Tests

- **stain.py**: fit on the reference patch itself → transform  $\approx$  identity (per-channel MAE < 3/255).
- Determinism: same slide fixture twice → byte-identical **stain\_params.json**.
- **qc\_checks.py**: unit tests with synthetic fixtures — gaussian-blurred patch flunks VoL, drawn green rectangle triggers pen detector, blank slide fails coverage.
- Golden verdicts on 3 checked-in small fixture slides.

## 6. Acceptance checklist

- ☐ Both dev slides (SVS + NDPI): stain params fitted, norm layer toggles correctly at  $\leq 10\times$ .
- ☐ A deliberately blurred copy of a fixture slide → QC fail with focus message; override path works and audits.
- ☐ Clean slide auto-advances (no unnecessary click — pathologist time is the budget).
- ☐ Re-run of preprocess on same slide produces identical stain params (determinism).
- ☐ QC calibration run on archive sample documented: sensitivity  $\geq 90\%$  hard-fails, false-fail  $\leq 2\%$ .